KUMMER THEORY AND THE FIRST QUESTIONS OF ANABELIAN GEOMETRY

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Abstract. The action of a Galois group on the roots of a nonzero field element defines a cocycle, and the resulting cohomology class records precisely its class modulo powers. We prove the Kummer isomorphism, including the finite Hilbert 90 argument needed for surjectivity, and calculate quadratic and cubic examples. Root torsors, finite covers of the three-punctured line, and recovery of a profinite group from its finite actions connect these calculations with fundamental groups. We then distinguish arithmetic from geometric fundamental groups and exhibit punctured curves whose geometric groups do not determine their algebraic structures. Standard foundational inputs from the theory of finite étale covers are identified explicitly; no deep anabelian reconstruction theorem is proved or assumed in the elementary calculations. Six exercises include complete solutions.

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1. The data: a field, a group, and a coefficient module

Let K be a field, fix a separable closure K^{sep}, and let $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$. This is a profinite group: its topology has as a basis of neighborhoods of the identity the subgroups fixing finite Galois extensions of K. Fix an integer $n \geq 2$ with $\mathrm{char}(K) \nmid n$, and put

$$\mu_n = \{\zeta \in (K^{\mathrm{sep}})^\times : \zeta^n = 1\}.$$

The polynomial $T^n - 1$ has distinct roots, since its derivative does not vanish at a root. It splits in K^{sep}, so μ_n has n elements. It carries the natural action of G_K and the discrete topology. This action must be specified: a coefficient group with a Galois action is more data than its underlying abstract group.

The elementary prerequisites are finite Galois theory, finite group actions, and polynomial factorization. We use the fixed-field theorem for Galois extensions, the fundamental theorem of finite Galois theory, and the fact that an automorphism of a finite normal extension extends to a separable closure. We prove the cohomological and categorical calculations used in the main examples. Later, the general classification of finite étale covers, the geometric–arithmetic exact sequence, and the comparison with complex topological covers will be stated as external foundational inputs with references. Those inputs are not elementary consequences of Kummer theory.

Our running question is narrower than full reconstruction of a field: *what does the action on the roots of $a \in K^\times$ determine about a?* The answer will be its class in $K^\times/(K^\times)^n$, not the original element, its sum with another element, or the entire ring structure of K.

2. Cocycles, coboundaries, and change of a chosen root

Definition 2.1. For a profinite group G acting continuously on a discrete abelian group M, written multiplicatively, a continuous 1-*cocycle* is a map $c : G \to M$ satisfying

$$c(gh) = c(g)\, g(c(h)). \tag{2.1}$$

For $\xi \in M$, the cocycle $b_\xi(g) = g(\xi)/\xi$ is a 1-*coboundary*. Define

$$H^1(G, M) = Z^1(G, M)/B^1(G, M),$$

where the numerator is the group of continuous cocycles under pointwise multiplication and the denominator is the subgroup of coboundaries.

To check the subgroup assertions, pointwise products of cocycles satisfy (2.1) because M is abelian; inverses satisfy the same identity. For a coboundary,

$$\frac{gh(\xi)}{\xi} = \frac{g(\xi)}{\xi}\, g\left(\frac{h(\xi)}{\xi}\right).$$

Also $b_{\xi\eta} = b_\xi b_\eta$, so these form a subgroup. Every cocycle has $c(1) = 1$, by applying (2.1) to $1 \cdot 1$. When the action on M is trivial, a cocycle is exactly a continuous group homomorphism, and every coboundary is trivial. Without this hypothesis a cocycle need not be a homomorphism.

For $a \in K^\times$, choose $\alpha \in K^{\mathrm{sep}}$ with $\alpha^n = a$ and set

$$c_a(g) = \frac{g(\alpha)}{\alpha}. \tag{2.2}$$

The choice is possible because $T^n - a$ is separable, and all its roots are obtained by multiplying one root by μ_n. Taking nth powers shows $c_a(g) \in \mu_n$. Direct calculation gives

$$\frac{gh(\alpha)}{\alpha} = \frac{g(\alpha)}{\alpha}\, g\left(\frac{h(\alpha)}{\alpha}\right),$$

so this is a cocycle. It is continuous: choose a finite Galois extension L/K in K^{sep} containing α and all of μ_n; the formula depends only on the restriction of g to L.

Replacing α by $\alpha' = \xi\alpha$, $\xi \in \mu_n$, gives

$$c'_a(g) = c_a(g)\frac{g(\xi)}{\xi}. \tag{2.3}$$

The representative can change, but its cohomology class does not. Replacing a by ab^n, $b \in K^\times$, and choosing the root $b\alpha$ leaves the representative unchanged. Finally, roots α and β for a, b give the root $\alpha\beta$ for ab, so $c_{ab} = c_a c_b$ with these choices. We have therefore constructed a well-defined group homomorphism

$$\kappa_{K,n} : K^\times/(K^\times)^n \longrightarrow H^1(G_K, \mu_n), \qquad [a] \longmapsto [c_a]. \tag{2.4}$$

3. The finite Hilbert 90 calculation

We give the ingredient needed to prove surjectivity of (2.4), rather than merely invoking a long exact sequence.

Lemma 3.1 (Independence of field automorphisms)**.** *Distinct automorphisms $g_1, \dots, g_m$ of a field L, regarded as functions $L \to L$, are linearly independent over L.*

Proof. Suppose a nonzero relation $\sum_{j=1}^{m} a_j g_j(t) = 0$ holds for all $t \in L$. Choose one with the smallest number of nonzero coefficients, and delete any zero terms. A one-term relation is impossible by evaluation at $t = 1$. For two distinct maps g_1, g_m, choose $x \in L$ with $g_1(x) \neq g_m(x)$. Apply the relation to xt and subtract $g_m(x)$ times the relation applied to t. This gives

$$\sum_{j=1}^{m-1} a_j\big(g_j(x) - g_m(x)\big)g_j(t) = 0 \quad \text{for all } t.$$

It has fewer terms and has a nonzero first coefficient, contradicting minimality. □

Theorem 3.2 (Finite Hilbert 90)**.** *Let L/K be finite Galois with group H. If $d : H \to L^\times$ satisfies $d(gh) = d(g)g(d(h))$, there is $\alpha \in L^\times$ such that*

$$d(h) = h(\alpha)/\alpha \quad (h \in H). \tag{3.1}$$

Proof. By Lemma 3.1, the function $t \mapsto \sum_{g\in H} d(g)g(t)$ is not identically zero. Choose t with $b = \sum_{g\in H} d(g)g(t) \neq 0$. Then

$$\begin{aligned} h(b) &= \sum_{g\in H} h(d(g))\, hg(t) = d(h)^{-1} \sum_{g\in H} d(hg)\, hg(t) \\ &= d(h)^{-1} b, \end{aligned}$$

since $g \mapsto hg$ permutes H. Taking $\alpha = b^{-1}$ gives $h(\alpha)/\alpha = b/h(b) = d(h)$, as required. □

The order of the quotient matters: our coboundary convention is $h(\alpha)/\alpha$, so the constructed element is b^{-1}, not b.

4. The Kummer isomorphism with complete proof

Theorem 4.1. *For $n \geq 2$ prime to $\mathrm{char}(K)$, the map (2.4) is an isomorphism:*

$$K^\times/(K^\times)^n \simeq H^1(G_K, \mu_n). \tag{4.1}$$

The G_K-action on μ_n need not be trivial.

Proof. Well-definedness and multiplicativity were established in Section 2. For injectivity, suppose $[c_a] = 0$. There is $\xi \in \mu_n$ with $g(\alpha)/\alpha = g(\xi)/\xi$ for all $g \in G_K$. Then α/ξ is fixed by G_K, hence lies in $K^\times$. Its nth power is a, showing $[a] = 1$.

For surjectivity, let $c : G_K \to \mu_n$ be a continuous cocycle. The open set $c^{-1}(1)$ contains the identity. Choose an open normal subgroup N contained in this set and in the kernel of the action on μ_n. Such a subgroup exists because open normal subgroups form a neighborhood basis in a profinite group. Then

$$c(gn) = c(g), \qquad c(ng) = c(g) \quad (n \in N),$$

the first equality using $c(n) = 1$ and the second also using the trivial action of N. Thus c descends to $H = G_K/N$. Let $L = (K^{\mathrm{sep}})^N$. It is a finite Galois extension with group H, contains μ_n, and the descended cocycle takes values in $\mu_n \subset L^\times$.

By Theorem 3.2, it has the form $h(\alpha)/\alpha$ for some $\alpha \in L^\times$. Since its values are nth roots of unity, $h(\alpha^n) = \alpha^n$ for all $h \in H$. Hence $a = \alpha^n \in K^\times$. The cocycle obtained from this root α is the original c, proving surjectivity. □

The familiar exact sequence behind the theorem is

$$1 \longrightarrow \mu_n \longrightarrow (K^{\mathrm{sep}})^\times \xrightarrow{z \mapsto z^n} (K^{\mathrm{sep}})^\times \longrightarrow 1.$$

Surjectivity holds because every $T^n - a$ with nonzero $a \in K^{\mathrm{sep}}$ is separable and has a root in the separably closed field. Its connecting map is exactly (2.2). The preceding proof makes explicit why the first cohomology of the middle multiplicative module supplies no further obstruction. The scheme-theoretic form is the Kummer exact sequence of étale sheaves when n is invertible [1, Tag 03PK].

If $K \subset F \subset K^{\mathrm{sep}}$ is a finite separable extension, restriction of cocycles along $G_F \subset G_K$ corresponds to the map on power classes induced by $K^\times \subset F^\times$. Indeed, one can use the same root α and the same formula on the subgroup. This proves naturality directly and shows how a nontrivial class can become trivial after extending the field.

5. Three explicit arithmetic examples

5.1. **Quadratic characters over the rationals.** For $K = \mathbb{Q}$ and $n = 2$, the coefficient group $\mu_2 = \{1, -1\}$ has trivial Galois action. Thus

$$H^1(G_{\mathbb{Q}}, \mu_2) = \mathrm{Hom}_{\mathrm{cont}}(G_{\mathbb{Q}}, \{\pm 1\}).$$

For $a = 2$, choose $\alpha = \sqrt{2}$. The character χ_2 has values 1 on automorphisms fixing $\sqrt{2}$ and -1 on those sending it to $-\sqrt{2}$. It factors through $\mathrm{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) \simeq \mathbb{Z}/2\mathbb{Z}$ and has kernel $G_{\mathbb{Q}(\sqrt{2})}$. For $a = 8$, choose $\sqrt{8} = 2\sqrt{2}$; the factor 2 is rational, so

$$\chi_8(g) = \frac{g(2\sqrt{2})}{2\sqrt{2}} = \chi_2(g).$$

This is exactly the equality $[8] = [2]$ because $8/2 = 2^2$. For $a = -1$, choose $\alpha = i$. Complex conjugation gives value -1. For $a = 2$ the same conjugation fixes the positive real root and gives value 1. Thus these two characters, and their square classes, are distinct.

Every nonzero rational has a unique form $a = (-1)^\varepsilon \prod_q q^{e_q}$, with integer exponents of finite support over rational primes q. Modulo squares only the sign and the parities of the e_q survive. Therefore

$$\mathbb{Q}^\times/(\mathbb{Q}^\times)^2 \simeq \{\pm 1\} \times \bigoplus_{q \text{ prime}} \mathbb{Z}/2\mathbb{Z}. \tag{5.1}$$

A rational is a square exactly when its sign is positive and all prime exponents are even. This follows by taking half of every exponent for the sufficiency direction, and by doubling exponents for necessity.

5.2. **A cubic cocycle which is not a character.** Let $K = \mathbb{Q}$, $n = 3$, $\alpha = \sqrt[3]{2}$ the real root, and $\zeta = e^{2\pi i/3}$. The splitting field is $L = \mathbb{Q}(\alpha, \zeta)$. The polynomial $T^3 - 2$ is irreducible: a reducible cubic over $\mathbb{Q}$ has a rational root, and the rational root test leaves only $\pm 1, \pm 2$, none a root. Thus $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$. This field is real, so it does not contain ζ; adjoining ζ has degree two and $[L : \mathbb{Q}] = 6$.

The Galois group is generated by

$$r(\alpha) = \zeta\alpha, \quad r(\zeta) = \zeta, \qquad s(\alpha) = \alpha, \quad s(\zeta) = \zeta^{-1},$$

with $r^3 = s^2 = 1$ and $srs = r^{-1}$. For existence of r, the tower formula gives $[L : \mathbb{Q}(\zeta)] = 3$, and its three embeddings over $\mathbb{Q}(\zeta)$ send α to the three roots; normality makes them automorphisms. The map s is complex conjugation. The displayed relations follow by applying both sides to α and ζ.

The root cocycle has $c(r) = \zeta$ and $c(s) = 1$, but

$$c(sr) = s(c(r)) = \zeta^{-1} \neq c(s)c(r) = \zeta.$$

It is a cocycle, not an ordinary homomorphism. Its full table is

g	1	r	r^2	s	sr	sr^2
$c(g)$	1	ζ	ζ^2	1	ζ^2	ζ

where products of automorphisms mean composition with the rightmost map acting first. Replacing α by $\zeta\alpha$ changes, for example, $c(s)$ from 1 to ζ, by (2.3), but does not change its class. Over $\mathbb{Q}(\zeta)$ the coefficient action becomes trivial, so restriction to $\mathrm{Gal}(L/\mathbb{Q}(\zeta)) = \langle r \rangle$ is a character.

5.3. **Square classes in an odd local field.** For an odd prime p, every $a \in \mathbb{Q}_p^\times$ is $p^k u$ with $k \in \mathbb{Z}$ and $u \in \mathbb{Z}_p^\times$. We claim that

$$a \text{ is a square in } \mathbb{Q}_p \iff k \text{ is even and } \overline{u} \text{ is a square in } \mathbb{F}_p^\times. \tag{5.2}$$

Necessity follows by taking the valuation and then reducing the unit factor of a square. For sufficiency choose $t_1 \in \mathbb{Z}$ whose nonzero residue squares to $\overline{u}$. Suppose $t_j^2 \equiv u \pmod{p^j}$. For $t_{j+1} = t_j + bp^j$, the condition at the next precision is

$$\frac{t_j^2 - u}{p^j} + 2t_j b \equiv 0 \pmod{p}.$$

Its coefficient $2t_j$ is a unit, so there is a unique next digit $b \in \{0, \ldots, p-1\}$. The sequence converges in $\mathbb{Z}_p$ to t with $t^2 = u$. Then $p^{k/2}t$ is a square root of a. This proves (5.2) by explicit lifting.

The parity of k and the residue square class consequently give an isomorphism

$$\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2 \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{F}_p^\times/(\mathbb{F}_p^\times)^2.$$

Surjectivity follows by lifting any nonzero residue; the kernel is exactly the squares by the criterion. The residue square-class group has two elements: squaring on $\mathbb{F}_p^\times$ has kernel $\{\pm 1\}$, so its image has index two. For $p = 5$, representatives of the four classes are $1, 2, 5, 10$. Kummer theory gives exactly four classes in $H^1(G_{\mathbb{Q}_5}, \mu_2)$; three are nontrivial quadratic characters.

6. Root torsors and what a choice of origin does

Definition 6.1. A *torsor* for a group M is a nonempty set on which M acts freely and transitively. Choosing a point identifies it with M, but the torsor itself does not come with a distinguished point.

The set $R_a = \{\alpha \in K^{\mathrm{sep}} : \alpha^n = a\}$ is a torsor for μ_n by multiplication. It also carries a Galois action compatible with the coefficient action:

$$g(\zeta\alpha) = g(\zeta)g(\alpha).$$

Choosing a root α identifies every point with $\zeta\alpha$, and the Galois action in these coordinates is

$$g(\zeta\alpha) = g(\zeta)c_a(g)\alpha.$$

The cocycle identity is exactly what makes these formulas into a group action. The change-of-origin formula is (2.3).

Proposition 6.2. *The torsors R_a, R_b are isomorphic as G_K-equivariant μ_n-torsors if and only if $b/a \in (K^\times)^n$.*

Proof. If $b/a = t^n$ with $t \in K^\times$, multiplication by t gives the required isomorphism. Conversely, let f be such an isomorphism, choose $\alpha \in R_a$, and put $\gamma = f(\alpha) \in R_b$. For every g, equivariance gives

$$g(\gamma) = f(g(\alpha)) = f(c_a(g)\alpha) = c_a(g)\gamma.$$

Hence $g(\gamma/\alpha) = \gamma/\alpha$, so $t = \gamma/\alpha \in K^\times$ and $b/a = t^n$. □

Without the Galois action, all these torsors are isomorphic after choosing roots, so that forgetting the action loses precisely the distinction just calculated. Without the μ_n-action, an equivariant map of underlying root sets need not record the same torsor structure. Both actions belong to the data of the proposition.

The root set is the geometric fiber of the finite algebra

$$A_a = K[T]/(T^n - a).$$

This algebra has dimension n with basis $1, T, \ldots, T^{n-1}$ and is finite étale: over K^{sep} it is a product of n copies of K^{sep}, since the polynomial has distinct roots. It need not be a field. For example, $\mathbb{Q}[T]/(T^2 - 1) \simeq \mathbb{Q} \times \mathbb{Q}$ by the Chinese remainder theorem, whereas $\mathbb{Q}[T]/(T^2 - 2)$ is the quadratic field $\mathbb{Q}(\sqrt{2})$. A root torsor is trivial when it has a K-rational point; this does not mean all roots lie in K unless $\mu_n \subset K$.

7. A Kummer cover of the three-punctured line

For this section let K have characteristic zero. Put

$$U = \mathbb{P}^1_K \setminus \{0, 1, \infty\}, \qquad R = K[t, t^{-1}, (t-1)^{-1}].$$

For $n \geq 2$ define the cover by the finite algebra

$$S = R[z]/(z^n - t), \qquad Y = \operatorname{Spec} S \longrightarrow U = \operatorname{Spec} R.$$

It is free of rank n over R, with basis $1, z, \dots, z^{n-1}$. Both t and z are units, and the derivative nz^{n-1} of its defining polynomial is a unit in S. The standard monic-polynomial criterion therefore makes this a finite étale map. One can also see the absence of ramification from

$$\Omega_{S/R} = S\,dz/(nz^{n-1}dz) = 0;$$

here freeness supplies flatness and the derivative criterion supplies the étale assertion. In these coordinates,

$$S \simeq K[z, z^{-1}, (z^n - 1)^{-1}]. \tag{7.1}$$

The isomorphism substitutes $t = z^n$; inverses of t and $t - 1$ become the displayed inverses, and the reverse substitution recovers the presentation. This is a domain, including after extension to an algebraic closure. Thus the cover is geometrically connected.

Over $\mathbb{C}$, its source is $\mathbb{P}^1_{\mathbb{C}} \setminus (\{0, \infty\} \cup \mu_n)$ and the map is $z \mapsto z^n$. The compactified map ramifies at 0 and ∞, but these points are absent from the cover of U. All n roots of 1 are also removed, because their images are the missing point 1. For $n = 2$, the source has the four punctures $0, 1, -1, \infty$.

Choose the basepoint $t_0 = 2$ and the root $\alpha = 2^{1/n} > 0$ over $\mathbb{C}$. The fiber consists of $\zeta_n^j \alpha$, $j \in \mathbb{Z}/n\mathbb{Z}$. Analytic continuation around a positive loop about 0 multiplies a root by ζ_n, whereas a loop about 1 fixes it. Indeed, along a path in $\mathbb{C}^\times$ an nth root changes by $e^{2\pi i w/n}$, where w is the winding number of the path about 0. A small loop about 1 has winding number zero about 0. Thus geometric monodromy on the labeled fiber has

$$\gamma_0 : j \mapsto j + 1, \qquad \gamma_1 : j \mapsto j. \tag{7.2}$$

For $K = \mathbb{Q}$, specialization at a rational point $t = a \neq 0, 1$ gives $\mathbb{Q}[z]/(z^n - a)$, exactly the root algebra of Section 6. For $n = 2$, the fibers at 2 and 8 yield isomorphic quadratic torsors, whereas the fiber at -1 yields a different one. The geometric monodromy calculation is independent of this rational specialization; the Galois action on the specialized fiber records its arithmetic.

8. Recovering a group from all its finite actions

This categorical calculation is elementary and fully provable. It also clarifies the role of a fiber functor.

Let G be a profinite group. Let Fin_G be the category of finite sets with continuous left G-action and equivariant maps. Let $\mathcal{U} : \mathrm{Fin}_G \to \mathrm{Fin}$ be the functor forgetting the action. A natural automorphism η of $\mathcal{U}$ is a family of permutations η_S satisfying $\eta_T f = f \eta_S$ for every equivariant map $f : S \to T$. The components η_S themselves are not required to be equivariant.

Theorem 8.1. *The map $G \to \mathrm{Aut}(\mathcal{U})$ sending g to the family $s \mapsto g \cdot s$ is an isomorphism of groups. With the topology of pointwise convergence on finite sets on the right, it is an isomorphism of topological groups.*

Proof. Each family is natural by equivariance of the maps f. First suppose G is finite. On its regular left action, the right translations $R_h : x \mapsto xh$ are equivariant. Naturality implies $\eta_G(xh) = \eta_G(x)h$. Taking $x = 1$ shows $\eta_G(h) = gh$ for $g = \eta_G(1)$. For any finite G-set S and $s \in S$, the map $f_s : G \to S$, $h \mapsto h \cdot s$, is equivariant. Applying naturality at 1 gives $\eta_S(s) = f_s(\eta_G(1)) = g \cdot s$. This proves existence and uniqueness of the element g.

For general profinite G, let N vary over open normal subgroups. Apply the finite argument to the full subcategory of actions factoring through G/N. The component on G/N is left multiplication by some $g_N \in G/N$. Naturality for the quotient maps $G/N' \to G/N$, when $N' \subseteq N$, shows that these elements are compatible. The defining

inverse-limit description $G \simeq \varprojlim_N G/N$ gives a unique $g \in G$ with these residues. Every finite continuous action factors through a finite quotient, since its action map to a finite permutation group has open normal kernel. Thus $\eta_S(s) = g \cdot s$ for every S. Conversely, all components determine all g_N, proving injectivity.

The action on each finite S is continuous, so the resulting map is continuous. Conversely, fixing the image of $1 \in G/N$ fixes g_N; these conditions give the open cosets of N, a basis for the topology of G. The inverse map is therefore continuous as well. □

The theorem reconstructs a *group* from a category together with its specified forgetful functor. It does not reconstruct a ring from a group. The example starts with all finite actions of an already specified group, so its elementary proof should not be mistaken for an anabelian reconstruction theorem about a variety.

For a connected scheme X and a geometric point $\overline{x}$, the analogous fiber functor is

$$F_{\overline{x}} : \mathrm{F\acute{E}t}_X \longrightarrow \mathrm{Fin}, \qquad (Y \to X) \longmapsto Y_{\overline{x}}.$$

Here $\mathrm{F\acute{E}t}_X$ is the category of finite étale covers, and the fiber means its finite set of geometric points. One defines $\pi_1^{\text{ét}}(X, \overline{x}) = \mathrm{Aut}(F_{\overline{x}})$ with its profinite topology. The fundamental theorem of Galois categories supplies the equivalence

$$\mathrm{F\acute{E}t}_X \simeq \mathrm{Fin}_{\pi_1^{\text{ét}}(X,\overline{x})}. \tag{8.1}$$

This equivalence, including the assertion that every finite action arises from a cover, is an external foundational theorem [1, Tag 0BND]; it is not proved by merely writing the automorphism-group definition. Changing the geometric basepoint identifies the groups up to an inner automorphism.

For $X = \operatorname{Spec} K$, the equivalence is the familiar finite Galois correspondence. A finite étale K-algebra is a finite product of finite separable field extensions, and its geometric fiber is its set of K-algebra maps to K^{sep}. A transitive finite G_K-set has the form G_K/H for an open subgroup H, and comes from the extension $(K^{\mathrm{sep}})^H$. Decomposing a finite set into orbits corresponds to the product decomposition of the algebra. This identifies $\pi_1^{\text{ét}}(\operatorname{Spec} K)$ with G_K.

9. Arithmetic fundamental groups and outer actions

Let K be a number field, fix an embedding $K \subset \mathbb{C}$ and an algebraic closure $\overline{K}$, and let X/K be a smooth geometrically connected curve. The geometric–arithmetic exact sequence is

$$1 \longrightarrow \pi_1^{\text{ét}}(X_{\overline{K}}, \overline{x}) \longrightarrow \pi_1^{\text{ét}}(X, \overline{x}) \longrightarrow G_K \longrightarrow 1. \tag{9.1}$$

This is a standard theorem for geometrically connected, quasi-compact, quasi-separated schemes over a field [1, Tag 0BTX]. Our curves satisfy these hypotheses. Its proof uses descent of finite étale covers and the Galois-category theorem, beyond the elementary calculations in this article.

The quotient describes arithmetic; the kernel describes covers after the field has been algebraically closed. It is not enough to replace the middle group by its kernel. In particular, (9.1) determines an outer action

$$G_K \longrightarrow \mathrm{Out}\big(\pi_1^{\text{ét}}(X_{\overline{K}}, \overline{x})\big)\,. \tag{9.2}$$

Here $\mathrm{Out}(N) = \mathrm{Aut}(N)/\mathrm{Inn}(N)$. The group-theoretic derivation is simple: for an extension $1 \to N \to E \to G \to 1$, a lift e of $g \in G$ acts on N by conjugation. Another lift ne changes that automorphism by conjugation by n, so its class in $\mathrm{Out}(N)$ is independent of the lift. Multiplication of lifts shows the classes define a homomorphism.

A K-rational point $x : \operatorname{Spec} K \to X$ gives a section of (9.1), after making compatible basepoint choices: functoriality gives $x_* : G_K \to \pi_1^{\text{ét}}(X)$ and the composite with the projection is the identity, since the composite scheme morphism is the identity. Changing paths changes the section by conjugation in the geometric group. A section determines an actual action and a semidirect-product description; without a chosen section the outer action is the natural description. A section is extra data, not a canonical splitting for every curve.

For $U = \mathbb{P}^1_K \setminus \{0, 1, \infty\}$, the rational point 2 supplies such a section. Pulling a finite cover back along this section restricts the arithmetic group action to the Galois action on its fiber at 2. For the cover $z^n = t$, that fiber action is exactly (2.2) for $a = 2$. This gives a direct relation between the Kummer calculation and the arithmetic fundamental group.

Over $\mathbb{C}$, the punctured line is $\mathbb{C} \setminus \{0, 1\}$ and its topological fundamental group is free on two generators. For a topological derivation, remove small disjoint discs about 0 and 1 and restrict to a large surrounding disc. Radial collars do not change the homotopy type; the resulting disc with two holes deformation retracts onto a connected graph with two independent cycles. Equivalently, van Kampen gives

$$\pi_1^{\text{top}}(U(\mathbb{C}), 2) = \langle \gamma_0, \gamma_1, \gamma_\infty \mid \gamma_0\gamma_1\gamma_\infty = 1 \rangle \simeq F_2.$$

Paths to the small puncture loops are chosen so that the displayed boundary relation holds. Eliminating γ_∞ leaves no relations on the other two generators. The Riemann existence theorem identifies finite topological covers with finite étale covers of the complex algebraic curve. Consequently

$$\pi_1^{\text{ét}}(U_{\mathbb{C}}) \simeq \widehat{F}_2, \tag{9.3}$$

where the hat denotes profinite completion. This comparison is another external theorem, stated in [2, Theorem 3.4]; it is not an identification of the discrete group F_2 with the profinite group.

10. A concrete limitation: groups do not record all moduli

The limitation can be seen without any deep theorem. For $\lambda \in \mathbb{C} \setminus \{0, 1\}$ put

$$X_\lambda = \mathbb{P}^1_{\mathbb{C}} \setminus \{0, 1, \infty, \lambda\}.$$

The same topological argument, now for a disc with three holes, gives $\pi_1^{\text{top}}(X_\lambda) \simeq F_3$ for every λ. The Riemann existence theorem consequently gives $\pi_1^{\text{ét}}(X_\lambda) \simeq \widehat{F}_3$ for every λ. Nevertheless the algebraic curves need not be isomorphic.

An algebraic isomorphism of these open curves induces mutually inverse rational maps of $\mathbb{P}^1$. Any rational function extends to a morphism $\mathbb{P}^1 \to \mathbb{P}^1$ by writing it as a coprime pair of homogeneous polynomials of the same degree; the pair has no common zero. The two extended maps are inverse because their composites equal the identity on the dense open curves, hence everywhere. Degrees multiply under composition of nonconstant maps of $\mathbb{P}^1$, so both have degree one. They are therefore Möbius transformations $z \mapsto (az + b)/(cz + d)$, $ad - bc \neq 0$, carrying the puncture sets onto each other.

With the first three punctures labeled and required to map respectively to $0, 1, \infty$, a Möbius transformation is the identity. Indeed, fixing 0 forces $b = 0$, fixing ∞ forces $c = 0$, and fixing 1 forces $a = d$. Thus the labeled curves distinguish every λ.

For unlabeled punctures, normalize any ordered triple of distinct points (a, b, c) to $(0, 1, \infty)$ using

$$M_{a,b,c}(z) = \frac{(z - a)(b - c)}{(z - c)(b - a)},$$

interpreted by limits when a point is ∞. Applying this to the fourth point gives the cross-ratio parameter. Permuting the four points yields precisely the six possible values

$$\lambda, \quad 1-\lambda, \quad \lambda^{-1}, \quad (1-\lambda)^{-1}, \quad \frac{\lambda}{\lambda-1}, \quad \frac{\lambda-1}{\lambda}. \tag{10.1}$$

To check completeness, the adjacent swaps of the first and second, second and third, and third and fourth points change the normalized parameter by $x \mapsto 1-x$, $x \mapsto x/(x-1)$, and $x \mapsto 1-x$, respectively. These swaps generate all permutations, and those substitutions preserve the displayed six-element set. Conversely, $x \mapsto 1-x$ and $x \mapsto 1/x$ arise from permutations and generate all six expressions. This also proves that any value in the list is realized by a suitable Möbius transformation of the puncture sets.

For $\lambda = 2$, the list reduces to $\{2, -1, 1/2\}$, which does not contain 3. Hence X_2 and X_3 are not algebraically isomorphic, although their geometric fundamental groups are isomorphic. Even recording the conjugacy classes of puncture loops does not recover the complex parameter: an orientation-preserving homeomorphism of the punctured spheres can carry corresponding labeled punctures and their peripheral loops onto each other. Such a homeomorphism can be constructed by moving finitely many points along paths and extending each move inside a small disc.

This is a limitation of the *geometric* group over an algebraically closed field. It does not refute arithmetic anabelian results: their input includes additional arithmetic structure, and their statements have field and curve hypotheses. Replacing that input by $\widehat{F}_3$ would omit precisely the data they study.

11. Anabelian questions, reconstruction, and the IUT connection

A smooth curve with projective completion of genus g and r geometric punctures is called *hyperbolic* when $2g-2+r > 0$. Both the three-punctured line and the four-punctured examples satisfy this condition. Hyperbolicity is a relevant hypothesis, not by itself a claim that the geometric group determines every curve.

An isomorphism of curves over a fixed number field induces an isomorphism of their arithmetic fundamental groups commuting with the maps to G_K, well-defined up to conjugation by the geometric group. The anabelian isomorphism question asks when this construction gives a bijection

$$\mathrm{Isom}_K(X,Y) \longrightarrow \mathrm{Isom}^{\mathrm{out}}_{G_K}\big(\pi_1^{\text{ét}}(X), \pi_1^{\text{ét}}(Y)\big).$$

The right-hand side here means continuous isomorphisms commuting with the projections to the fixed G_K, modulo conjugation by the target geometric group. Surjectivity would mean that every permitted group isomorphism comes from geometry; injectivity would mean that geometry is distinguished by its action on covers. We formulate the question to locate the elementary examples, not as a theorem proved in this article.

The further distinction between comparison and reconstruction is important. In a comparison problem, two geometric objects and their groups are already present. A reconstruction problem instead seeks objects and operations constructed in language depending only on the specified group-theoretic input, with compatibility under its isomorphisms. Hoshi's introduction distinguishes these bi-anabelian and mono-anabelian viewpoints [3, Introduction]. His Theorem 2.2 also gives examples of nonisomorphic mixed-characteristic local fields with isomorphic absolute Galois groups, so an unrestricted claim that G_K always determines K would be false. This is a cited external result, not a construction or proof supplied here.

Our elementary conclusions are exact but narrower. Kummer theory identifies a multiplicative power class with a cohomology class, using both a group and its coefficient

module. The finite-action calculation recovers a group from its category with a fiber functor. The arithmetic exact sequence supplies a group together with its quotient and an outer action. None of these observations alone recovers the addition law of a field from a bare abstract group.

In Mochizuki's overview, Kummer theory enters the passage between multiplicative structures and the étale-theoretic descriptions used in IUT [4, §3, printed pp. 33–35]. Hoshi's discussion of cyclotomic synchronization and transport explains why identifications of coefficient modules are part of the problem [3, §§6–7]. In our cubic example, simply discarding the action on μ_3 already changes a cocycle problem into a wrong character problem. This is an elementary illustration of why those coefficient data cannot be suppressed.

The finite-level isomorphism (4.1) does not construct the IUT Kummer-detachment procedure, establish the required cyclotomic rigidity, or prove compatibility of the log- and Θ-links. Those constructions require substantial additional theory. The useful reading discipline supplied by this article is to identify the group, its topology, the coefficient action, the torsor or cover, and the equivalence relation on choices before asserting that one description recovers another.

12. Exercises with complete solutions

Exercise 1. For $K = \mathbb{Q}$ and $n = 2$, decide which of $2, 8, 18, 32, -2$ have the same Kummer class as 2. Describe the field over which that class becomes trivial.

Solution. Divide each element by 2. The ratios are respectively $1, 4, 9, 16, -1$. The first four are rational squares, whereas -1 is not a square in $\mathbb{Q}$. Thus $2, 8, 18, 32$ give the same class and -2 does not. Over $\mathbb{Q}(\sqrt{2})$ the class becomes trivial, since 2 is a square there. More generally it becomes trivial over an intermediate field $F \subset \mathbb{Q}^{\mathrm{sep}}$ exactly when $\sqrt{2} \in F$, since a square root of 2 must be $\pm\sqrt{2}$. □

Exercise 2. For the cubic example, replace α by $\alpha' = \zeta\alpha$. Calculate the new values at r, s, and verify the cocycle identity at (s, r) for both choices.

Solution. By (2.3), $c'(r) = \zeta\, r(\zeta)/\zeta = \zeta$ and $c'(s) = \zeta^{-1}/\zeta = \zeta^{-2} = \zeta$. For the original root, $c(sr) = \zeta^{-1} = c(s)s(c(r))$. For the new root,

$$c'(sr) = c(sr)\frac{sr(\zeta)}{\zeta} = \zeta^{-1}\frac{\zeta^{-1}}{\zeta} = 1,$$

while $c'(s)s(c'(r)) = \zeta\zeta^{-1} = 1$. The two cocycles differ by the coboundary of ζ and represent the same class, although their values at s differ. □

Exercise 3. Construct a $G_{\mathbb{Q}}$-equivariant bijection between the root sets R_2 and R_4 for $n = 3$. Show that it is not μ_3-equivariant, and explain why this does not contradict Kummer theory.

Solution. The map $f(\rho) = \rho^2$ sends $\rho^3 = 2$ to a root of 4. Its inverse on R_4 is $\eta \mapsto \eta^2/2$, since $(\eta^2/2)^3 = 16/8 = 2$ and $(\rho^2)^2/2 = \rho^4/2 = \rho$. Both formulas commute with $G_{\mathbb{Q}}$ because their coefficients are rational. But $f(\zeta\rho) = \zeta^2 f(\rho)$, not $\zeta f(\rho)$ for a primitive cube root ζ. Thus this is not an isomorphism of the torsors with their specified μ_3-actions. The ratio $4/2 = 2$ is not a rational cube, so Proposition 6.2 rules out such a torsor isomorphism. An isomorphism of plain Galois sets preserves less data. □

Exercise 4. Determine the square classes of $6, 8, 15, 20$ in $\mathbb{Q}_5^\times$, using the representatives $1, 2, 5, 10$. Which are squares?

Solution. The nonzero squares modulo 5 are $1, 4$. The number 6 has valuation zero and residue 1, so it has class 1 and is a square in $\mathbb{Q}_5$. The number 8 has valuation zero and nonsquare residue 3; since $8/2 = 4$ is already a rational square, its class is 2. For $15 = 5 \cdot 3$, the valuation is odd and the unit residue is nonsquare, so its class is 10. Finally, $20 = 5 \cdot 4$ has odd valuation and square unit residue, so its class is 5. Only 6 among the four numbers is a square in $\mathbb{Q}_5$. Its square root need not be rational; the digit lifting in (5.2) constructs it in $\mathbb{Z}_5$. □

Exercise 5. For the cover $z^n = t$ of $U_{\mathbb{C}}$, determine the image and kernel of the monodromy homomorphism $F_2 \to \mathbb{Z}/n\mathbb{Z}$. Compute the Euler characteristics of its source and target.

Solution. By (7.2), γ_0 maps to 1 and γ_1 to 0, so the image is all of $\mathbb{Z}/n\mathbb{Z}$. For a word in the two generators, the image is the total exponent of γ_0 modulo n. The kernel is therefore the words with that exponent divisible by n; it is also the normal subgroup generated by γ_1 and γ_0^n, since imposing these relations leaves exactly the cyclic group of order n. The target has three punctures and Euler characteristic $2 - 3 = -1$; the source has $n + 2$ punctures by (7.1), so its Euler characteristic is $2 - (n + 2) = -n$. This agrees with multiplication of Euler characteristic by the degree n of an unbranched finite cover, as can be seen by lifting the cells of a finite graph model. □

Exercise 6. Give an explicit isomorphism between X_2 and X_{-1} from Section 10. Explain why their common geometric fundamental group cannot distinguish X_2 from X_3, whereas the cross-ratio calculation does.

Solution. The Möbius transformation $z \mapsto 1 - z$ sends the puncture set $\{0, 1, \infty, 2\}$ to $\{0, 1, \infty, -1\}$, so restricts to an algebraic isomorphism $X_2 \to X_{-1}$. It swaps the punctures $0, 1$, so it does not preserve the ordered marking of those two points. All four-punctured spheres have topological group F_3, and their geometric étale groups are all $\widehat{F_3}$. This abstract group therefore cannot distinguish X_2 from X_3. In contrast, the cross-ratio orbit of 2 is $\{2, -1, 1/2\}$, excluding 3. Every algebraic isomorphism would give a Möbius transformation of the puncture sets and force membership in that orbit, so no such isomorphism exists. Arithmetic reconstruction questions require additional input beyond this geometric group. □

References

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